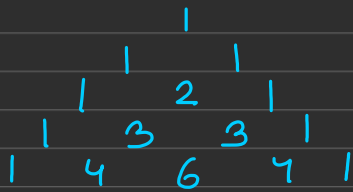


Pascals Triangle:-



We know, $nCr = \frac{n!}{r!(n-r)!} \Rightarrow \overset{\rightarrow n}{5} \underset{\rightarrow r}{2} \Rightarrow \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times (3 \times 1)} \Rightarrow \frac{20}{2} \Rightarrow 10$

$\rightarrow$ always take the first 'r' digits, as others will cancel out

Now,

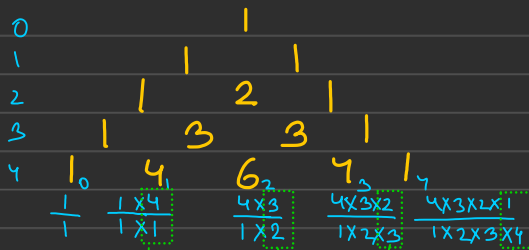
for i^{th} row and j^{th} col, print the value

Pseudo code:-

```
for (int i=0 → n)
    new list < int >,
    for (j=0 → i+1)
        list.add(nCr(i, j)),
    res.add(list),
```

1 → 0th row, 1 el
1 1 → 1st row, 2 el
1 2 1 → 2nd row, 3 el

val = 1
for (int i=0 → n)
 val = (val × (n-i)) / (i+1),



Something to the power no
is getting multiplied

num = row - col + 1
deno = col

Pseudo code:-

```
for (int i=0 → n)
    new list < int >,
    list.add(1),
    for (j=0 → i+1)
        list.add((i-j+1)/j * list.get(i)),
    res.add(list),
```